TAO Wenwen

Curriculum Vitae — PhD and Graduate Admissions

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2027 PhD · Research Master · MPhil Applicant | Full Academic CV

B.M. Clinical Medicine, Nanchang University (2017–2022) | Chinese (Native) | English (IELTS 6.5, TOEFL 97)

Research Summary

Clinically trained researcher working at the intersection of **translational neuroscience, computational psychiatry and multimodal clinical AI**. My research spans psychiatric cohorts in addiction and depression, multimodal neuroimaging and biosignals (MRI/DTI, EEG–fNIRS, eye tracking and physiology), neuromodulation (rTMS, TUS), behavioral phenotyping, microbiome/metabolomics and experimental-system development.

My current research focuses on **decoding dynamic brain and behavioral states and translating them into clinically meaningful, machine-actionable representations**. I am particularly interested in transforming standardized psychiatric frameworks such as **K-SADS-PL DSM-5, CDRS-R and MINI** into structured multimodal data specifications, age-adaptive dialogue workflows and evaluation frameworks for AI-enabled mental-health assessment and human–AI interaction.

Across my recent work, I have increasingly taken an end-to-end translational role—linking **clinical protocol design → multimodal acquisition and data governance → state representation and AI input/output specification → deployment-aware system architecture**. This includes designing research systems that account from the outset for synchronization, data quality, interpretability, edge–cloud computation, latency and safety constraints.

My longer-term goal is to investigate how **internal neural, physiological and behavioral states can be decoded, modeled and translated into adaptive actions** in clinical and human–AI systems.

Research Interests

Multimodal AI for Mental Health · Computational Psychiatry · Dynamic Brain-State Decoding · Multimodal Neuroimaging · Clinical AI · LLM/MLLM-Based Mental Health Assessment · Human–AI Interaction and Dialogue Systems · Neuromodulation · Translational Neuroscience · Medical Knowledge Engineering · Intelligence-to-Action Interfaces

Education

2017–2022 | Nanchang University

B.M. in Clinical Medicine

GPA: **3.2/4.0 (82/100)**

Relevant training: Psychiatry, Neurology, Internal Medicine, Surgery, Pathology, Pharmacology, Medical Statistics and Evidence-Based Medicine

Research Experience

MODULE 1 — Multimodal AI for Clinical Psychiatry

Mar 2026–Present | Tsinghua University — CogDrive Lab

State Key Laboratory of Intelligent Green Vehicle and Mobility

Research Assistant | Advisor: Prof. Dongpu Cao

Collaborating Hospital: Tongji Hospital, Huazhong University of Science and Technology

Project: Multimodal AI for Adolescent Depression Screening and Human–AI Interaction in Intelligent Cockpits

Project Context: Multimodal clinical-AI translation for intelligent health-cockpit research, connecting laboratory-based psychiatric phenotyping with MLLM-enabled state assessment, dialogue interaction and deployment-aware in-vehicle systems.

- **Independently conceived and led a prospective multimodal clinical cohort study** in children and adolescents aged 6–18, with planned enrollment of **300 clinical participants and 150 healthy controls**. Introduced an education- and cognitive-development–stratified design across four age bands (6–9, 9–12, 12–15 and 15–18 years), tailoring assessment procedures to age-specific cognitive abilities, psychosocial stressors and psychiatric presentations. The protocol underwent joint review by the Tsinghua research team and Tongji Hospital clinical collaborators.
- **Designed an end-to-end synchronized multimodal acquisition framework**, integrating **64-channel EEG–fNIRS, desktop eye tracking, multi-view facial video, speech and multi-channel physiological signals (ECG, PPG, GSR and respiration)** during both computerized paradigms and clinician-led interviews. Defined paradigm structure, acquisition architecture, event/trigger conventions, operator procedures, quality-control workflows and analysis-ready data specifications to support multimodal state modeling and exploratory EEG–fNIRS neurovascular-coupling analyses.
- **Established the study's clinical ground-truth framework** through a four-stage assessment pipeline: **CDRS-R → age-matched self-/parent-report measures → K-SADS-PL DSM-5 clinician confirmation → multisource severity grading**. Further mapped item-level CDRS-R clinical anchors to observable multimodal behavioral features, including **facial action units, speech prosody, gaze patterns and saccadic behavior**, creating interpretable links between psychiatric phenotypes and machine-readable signals.
- **Operationalized K-SADS-PL DSM-5 into four age-adaptive structured dialogue protocols** for ages 6–9, 9–12, 12–15 and 15–18, incorporating multilevel dynamic branching across seven clinical domains: mood, ADHD/externalizing behavior, somatic

symptoms, cognitive impairment, rejection sensitivity, self-esteem/hopelessness and escalating suicide-risk assessment. Designed this framework as the **core clinical logic layer for a speech-based multimodal large language model (MLLM)**, with planned downstream deployment on the **Qualcomm SA8397P automotive computing platform**.

- **Defined synchronization, metadata and quality-control standards for 13+ multimodal data streams**, specifying target temporal resolution at the data (microsecond), feature (millisecond) and semantic (~100-ms) layers, together with standardized metadata conventions, QC procedures and graded data-quality labels. The resulting specification is designed to generate model-ready datasets for downstream multimodal training and knowledge/model distillation; real-world on-site acquisition remains in the pre-collection hardware and system-validation stage.
- **Developed BPlus MVP-0 as the study's experimental operations platform**, implementing six modules—**Rest, MMN, VFT, N-back, Emotion and Recovery**—together with participant-response capture, event/trigger logging, preflight checks and automated QC/state reports. Software demonstrations and dry runs passed, establishing the workflow for subsequent real-hardware synchronization validation and pilot data collection.
- **Independently coordinated and advanced the full Tongji Hospital IRB approval process**, translating the experimental protocol into ethics-ready procedures for minor protection, informed consent/assent and stigma-sensitive recruitment. The study has received **IRB approval**, establishing the regulatory foundation for subsequent clinical data collection.
- **Led the product and system architecture design for the intelligent health-cockpit translation pathway**, linking laboratory-derived multimodal biomarkers with low-interference in-vehicle state monitoring. Designed a **Unified LLM orchestration architecture spanning six capability domains and 34 functions**, integrating specialist models, Agent/Skill orchestration, Function Calling, Tool/API/RAG access, structured result validation, confidence-aware routing, an **L0–L5 Safety Gate**, feedback/memory mechanisms and edge–cloud coordination.
- Served as a **translational bridge between clinical psychiatry and AI systems engineering**, converting clinical requirements into multimodal data specifications and MLLM input/output definitions while incorporating vehicle-side compute, real-time latency and safety constraints from the experimental-design stage.

MODULE 2 — Neuroimaging & Neuromodulation

Sep 2025–Mar 2026 | Tsinghua University — Yan Lab & Beijing Normal University — Liu Lab

Research Intern / Visiting Research Intern | Advisors: Prof. Chaogan Yan, Dr. Zheyun Liu

Depression rTMS Research and Parkinson's Disease TUS Research

Project 1: Mind Flower — Clinical Translation of rTMS Neuromodulation and Mindfulness-Based Counseling for Depression

Funding: NSFC / National Key R&D Program / STI2030 Brain Science / CAS (multi-phase)

- **Investigated multimodal and longitudinal markers of depression and rTMS treatment response** across a multi-phase clinical study, screening participants and independently administering **MINI, HAMD, HAMA and MADRS** assessments at baseline, intervention and follow-up. Converted clinical interviews into structured longitudinal records and maintained an access-controlled, NAS-based research database.
- **Managed multimodal MRI acquisition for resting-state, rumination and distraction paradigms**, preparing individualized negative-recall cues, adapting E-Prime tasks to participant-specific materials, monitoring head motion (**<3 mm**) and conducting OptiMRI think-aloud audio QC to support analysis of neural correlates and potential predictors of treatment response.
- **Tracked repeated rTMS intervention and cognitive-behavioral phenotypes**, documenting daily **HAMD-6** scores, stimulation sessions and adverse effects while administering **WCST, SST and emotional n-back** paradigms.
- Coordinated mindfulness-based counseling recordings and integrated laboratory measures with **two-week wearable monitoring of HRV, sleep and activity**, extending phenotyping from controlled experiments to longitudinal real-world signals.
- **Coordinated multi-month experimental workflows across clinicians, counselors and participants**, standardizing E-Prime data recovery and export, multimodal file naming, QC and hierarchical NAS archiving to improve reproducibility across repeated assessments and follow-up sessions.

Project 2: Transcranial Ultrasound Stimulation of the GPi for Cognitive Modulation in Parkinson's Disease

- **Supported MRI-guided transcranial ultrasound stimulation (TUS) targeting the bilateral globus pallidus internus (GPi)** in participants with Parkinson's disease, using structural MRI and **Brainsight neuronavigation** for individualized targeting and documenting stimulation parameters, including pulse-repetition frequency and duty cycle.
- **Operated a combined TUS–cognitive assessment protocol**, coordinating approximately **1.5 hours of memory and cognitive-flexibility testing** while recording behavioral responses and monitoring participant-reported auditory, thermal and other tolerability signals.
- The work supports investigation of whether **non-invasive deep-brain ultrasound neuromodulation can modulate basal-ganglia-related cognitive function** while maintaining standardized safety, parameter-recording and acquisition procedures.

MODULE 3 — Addiction Neuroscience & Translational Research

Jan 2024–Aug 2025 | Shenzhen Mental Health Center — Addiction Medicine

Research Assistant (Full-time) | Advisors: Prof. Mei Yang, Prof. Xiaojian Jia

Core contributor across parallel funded projects in addiction neuroscience, gut–brain mechanisms and multimodal neuroimaging

- **Investigated clinical, neural and biological heterogeneity across alcohol use disorder (AUD), methamphetamine dependence and matched healthy controls**, independently managing **290+ participants** and building a standardized multimodal research database integrating longitudinal clinical phenotypes, MRI/DTI, EEG, neurocognitive assessments, **16S rRNA microbiome sequencing and GC–MS short-chain fatty acid (SCFA) metabolomics**, with **>95% data completeness**.

P1. Gut Microbiota–Metabolite–Circuit Mechanisms in AUD

Shenzhen Basic Research Fund | 220 participants

- Investigated a putative **gut–metabolite–white-matter pathway** by integrating 16S rRNA microbiome profiles, plasma SCFAs and multimodal MRI.
- Identified **three candidate microbial markers associated with SCFA levels** and observed associations between lower SCFA concentrations and reduced fractional anisotropy (FA) in the **anterior limb of the internal capsule and corpus callosum**, providing preliminary evidence linking gut-derived metabolic alterations to white-matter microstructure.

P2. Neuroprotective Mechanisms of Tetramethylpyrazine (TMP) in AUD

Shenzhen Science & Technology Innovation Commission | 160 participants

- Coordinated multimodal MRI acquisition and evaluated pre-/post-intervention FA changes in the **corpus callosum and superior longitudinal fasciculus**.
- Preliminary imaging differences require longitudinal confirmation; the project also produced an optimized **DTI + resting-state fMRI acquisition and QC workflow**, including procedures to reduce susceptibility-related artifacts in subcortical regions.

P3. Methamphetamine-Related Frontotemporal Injury and Memory Impairment

Guangdong Natural Science Foundation | 10 deeply phenotyped participants + 80-case database

- Independently acquired MRI and EEG data and examined **prefrontal–hippocampal structural pathways, cingulum and fornix diffusion measures, and structural–functional coupling** using DTI and fMRI.
- Contributed to the characterization of distributed frontotemporal circuit abnormalities associated with chronic methamphetamine exposure.

P4. Prefrontal–Ventral Striatal Executive-Control Network Impairment

Shanghai Mental Health Center Collaboration | 20 deeply phenotyped cases

- Examined whether **PFC–ventral striatal structural connectivity alterations were associated with executive-network dysfunction and relapse vulnerability**, integrating DTI, task-based fMRI and behavioral measures.
- Developed an **exploratory logistic-regression model of relapse outcome** and generated connectivity-based hypotheses for future individualized neuromodulation research.

P5. Cross-Project Data Infrastructure and Harmonization

- **Designed the shared data infrastructure supporting all four projects**, including sample-type and longitudinal time-point labels, clinical-severity strata, data-quality grades, anomaly annotations and a unified **Data Dictionary**.
- Established end-to-end SOPs spanning biospecimen collection, data coding, three-tier QC, anomaly handling, version control and archiving.
- Developed harmonization rules addressing differences in MRI acquisition parameters, assessment versions and biospecimen-processing protocols across collaborating centers, supporting reproducible cross-project and multicenter analyses.

Selected Research Contributions

- **Clinical AI and adolescent mental health:** Designed a prospective **450-participant** multimodal clinical framework combining **64-channel EEG–fNIRS, eye tracking, facial video, speech and physiology**, with age-adaptive psychiatric dialogue, standardized clinical ground truth and deployment-oriented MLLM specifications.
- **Intelligent health-cockpit architecture:** Designed a **Unified LLM–centered architecture spanning six capability domains and 34 functions**, connecting multimodal perception, health assessment, dialogue, safety, cabin control and data/knowledge management through Agent/Skill orchestration, Function Calling, validation and feedback mechanisms.
- **Gut–brain mechanisms in AUD:** Integrated **16S rRNA, GC–MS SCFAs and multimodal MRI in 220 participants**, identifying candidate microbial–metabolite associations and corresponding variation in white-matter microstructural measures.
- **Addiction neuroimaging:** Coordinated a **160-participant TMP imaging study**, characterized methamphetamine-related frontotemporal circuitry, and examined **PFC–ventral striatal structural–functional relationships** in deeply phenotyped addiction cohorts.
- **Neuromodulation and longitudinal phenotyping:** Contributed to multimodal **rTMS research in depression** and MRI-guided bilateral **GPi TUS research in Parkinson's disease**, integrating clinical scales, behavioral paradigms, MRI, wearables and safety monitoring.
- **Research infrastructure:** Developed multimodal synchronization, QC, metadata, longitudinal labeling, Data Dictionary and multicenter harmonization frameworks spanning clinical psychiatry, neuroimaging, biosignals and molecular data.

Publications / Manuscripts

- **Tao, W. Early Depressive Symptom Severity and Six-Month Return to Drinking After Inpatient Treatment for Alcohol Dependence: A Prospective Cohort Study.** Manuscript submitted to *Psychiatry Research*.
- **Tao, W. Temporal Dataset Shift Limits Relapse-Risk Stratification Across the Inpatient-to-Community Transition in Alcohol Dependence: A Temporal Comparison of Conventional Machine Learning and Self-Supervised Representation Learning.** Submitted to *Addiction Science & Clinical Practice (ASCP)*.
- **Tao, W. Task state dominates participant-held-out prediction of risky decisions in methamphetamine dependence, with limited incremental value from pre-decision EEG.** Submitted to *Clinical EEG and Neuroscience*.

Clinical Experience

Dec 2022–Dec 2023 | Wannian County Hospital of TCM

Resident Physician

- Managed **15–20 chronic-disease inpatients** through clinical assessment, treatment planning, inpatient management and follow-up, gaining full-cycle experience in routine clinical care.

- Developed standardized medical-record templates to improve the consistency, traceability and retrieval of longitudinal clinical information.

Feb 2021–Feb 2022 | Ganzhou People's Hospital

Clinical Intern

- Completed **12 clinical rotations**, including Neurology ICU and Cardiology ICU, with broad exposure to inpatient, emergency and perioperative care.
- Performed supervised clinical procedures including **lumbar puncture and ECG interpretation**, and participated in routine surgical and inpatient workflows.

Methods and Technical Skills

Clinical Research & Psychiatric Assessment

MINI; K-SADS-PL DSM-5; CDRS-R; HAMD; HAMA; MADRS; PHQ-9; PHQ-15; CDI; SDQ; clinical protocol development; longitudinal cohort design; IRB coordination; informed consent/assent; pediatric research safeguards

Neuroimaging & Connectivity Analysis

Structural MRI; resting-state and task-based fMRI; DTI tractography; FA/ADC analysis; structural–functional connectivity; DPABI; DPABI-Surf; FSL; individualized MRI-based targeting

Neurophysiology & Multimodal Sensing

EEG (**32/64-channel**); simultaneous **EEG–fNIRS**; ECG; PPG; GSR; respiration; desktop eye tracking; synchronized task/interview acquisition; event-trigger design; multimodal temporal alignment; signal QC; exploratory neurovascular-coupling analysis

Neuromodulation

rTMS; transcranial ultrasound stimulation (**TUS**); MRI-based **Brainsight neuronavigation**; individualized targeting; stimulation-parameter logging; tolerability and safety monitoring

Behavioral & Cognitive Paradigms

MMN; VFT; N-back; emotional N-back; WCST; SST; BART; Two-Step task; clinical interview paradigms; E-Prime and PsychoPy experimental workflows

Microbiome, Metabolomics & Biospecimens

16S rRNA gut-microbiome sequencing; GC–MS short-chain fatty acid metabolomics; blood/stool/saliva processing; biospecimen collection, labeling and QC SOPs

Programming, Statistics & Clinical AI

Python; R; SPSS; MATLAB; statistical modeling; logistic regression; multimodal data schemas; age-adaptive clinical dialogue workflows; LLM/MLLM input/output specifications; prompt and workflow design for clinical AI; RAG; edge–cloud system architecture

Research Software & Data Infrastructure

BPlus MVP-0 experimental platform; participant-response and event logging; trigger architecture; automated preflight and QC reporting; longitudinal database design; Data Dictionary development; version control; NAS-based data management; multicenter data harmonization